Unit 6: Emerging Technologies

Complete exercise solutions — MCQs, Short Questions, Long Questions

EXERCISE SOLUTIONS — Unit 6: Emerging Technologies

Multiple Choice Questions

1. The main benefit of edge computing is:

• a) Lower cost

• b) Reduced latency

• c) Increased complexity

• d) Enhanced security

Answer: b) Reduced latency

2. A cloud deployment model with resources shared among multiple organizations with common concerns is:

• a) Public Cloud

• b) Private Cloud

• c) Community Cloud

• d) Hybrid Cloud

Answer: c) Community Cloud

3. The advantage of using a distributed ledger in blockchain technology is:

• a) Centralized control for quick decision-making

• b) Easy alteration of transaction histories

• c) Enhanced transparency and security through decentralized verification

• d) Lower computational requirements

Answer: c) Enhanced transparency and security through decentralized verification

4. A cloud deployment model combining public and private cloud features is:

• a) Public Cloud

• b) Hybrid Cloud

• c) Community Cloud

• d) Multi-Cloud

Answer: b) Hybrid Cloud

5. The purpose of a distributed ledger in blockchain is:

• a) Central authority management

• b) Secure and transparent data sharing among multiple participants

• c) Fewer participants required

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• d) Data visibility only to central authority

Answer: b) Secure and transparent data sharing among multiple participants

6. A cloud service offering a platform for developing, running, and managing applications without managing infrastructure is:

• a) Infrastructure as a Service (IaaS)

• b) Platform as a Service (PaaS)

• c) Software as a Service (SaaS)

• d) Data as a Service (DaaS)

Answer: b) Platform as a Service (PaaS)

7. The service model enabling application deployment without server management is:

• a) Infrastructure as a Service (IaaS)

• b) Platform as a Service (PaaS)

• c) Software as a Service (SaaS)

• d) Serverless Architecture

Answer: d) Serverless Architecture

8. The feature introduced in Blockchain 2.0 beyond cryptocurrency is:

• a) Enhanced mining techniques

• b) Decentralized applications and smart contracts

• c) Better graphics

• d) Faster internet speeds

Answer: b) Decentralized applications and smart contracts

9. The primary advantage of serverless architectures is:

• a) Cost savings

• b) Constant server management

• c) Increased hardware needs

• d) Manual scaling

Answer: a) Cost savings

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Short Questions

1. Analyze the role of Peer-to-Peer Networks in Blockchain. How do they function and why are they essential?

In a blockchain, peer-to-peer (P2P) networks connect all participating computers (nodes) directly, without a central server. Each node stores a copy of the ledger and validates transactions. When a new transaction occurs, it is broadcast to all nodes, which verify it through consensus mechanisms before adding it to the blockchain. P2P networks are essential because they provide decentralisation (no single point of failure), transparency (everyone sees the same ledger), and security (altering records would require controlling most nodes).

2. Describe the concept of immutability in blockchain. Why is it a critical feature?

Immutability means that once a block is added to the blockchain, its data cannot be altered or deleted. Each block contains a cryptographic hash of the previous block, so changing any record would break the chain and be detected by all nodes. It is critical because it ensures trust and integrity — transaction histories are permanent and tamper-proof, which is essential for applications like financial records, supply chain tracking, and legal documents.

3. What is edge computing and how does it benefit data processing?

Edge computing processes data close to its source (at the 'edge' of the network) rather than in a centralised data centre. It benefits data processing by: (1) reducing latency — data does not travel far, so responses are faster; (2) saving bandwidth — only important results are sent to the cloud; (3) improving reliability — the system works even if the central network fails. Example: in autonomous vehicles, sensor data is processed locally in the car for quick real-time decisions.

4. Describe the concept of serverless architectures.

Serverless architecture is a cloud computing model where the cloud provider manages the servers, and developers only write and deploy functions (small pieces of code) that run in response to events. There is no server to provision or maintain. Example: AWS Lambda runs a function whenever a user uploads a photo. The provider handles scaling, patching, and infrastructure. Developers pay only for the time their code actually runs.

5. What advantages do serverless architectures offer to developers?

Serverless architectures offer: (1) Cost savings — you pay only for execution time, not for idle servers. (2) No server management — no need to provision, patch, or scale servers. (3) Automatic scaling — the platform scales functions up or down based on demand. (4) Faster development — developers focus on code, not infrastructure. (5) High availability — the provider ensures reliability. They are ideal for event-driven applications like image processing, APIs, and data pipelines.

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6. How does edge computing improve the efficiency of autonomous vehicles?

Autonomous vehicles generate huge amounts of sensor and camera data that must be processed in milliseconds to make safe decisions. Edge computing processes this data locally in the vehicle instead of sending it to a distant data centre. This reduces latency (no round-trip delay), enables real-time responses to road conditions, saves bandwidth, and keeps the vehicle functional even without internet connectivity. Without edge computing, the delay of sending data to the cloud and back would be dangerous.

7. Differentiate between Elasticity and On-Demand access in cloud computing.

Elasticity is the cloud's ability to automatically scale resources up or down based on real-time demand. For example, an e-commerce site adds servers during a sale and removes them afterward. On-demand access means users can provision computing resources (servers, storage) whenever they need them, without buying hardware or waiting for setup. In short: on-demand access is about getting resources when you need them; elasticity is about the resources automatically adjusting to match demand.

Long Questions

1. Define cloud deployment models and assess the differences among them.

Cloud deployment models describe how cloud services are delivered: (1) Public Cloud — services over the public internet, shared by many organisations (e.g., AWS, Google Cloud). Cheap and scalable but less control. (2) Private Cloud — services dedicated to a single organisation, hosted internally or by a third party. More secure and controlled but more expensive. (3) Hybrid Cloud — combines public and private clouds, allowing data and applications to move between them. Flexible but complex. (4) Community Cloud — shared by organisations with common concerns (e.g., hospitals, government agencies). (5) Multi-Cloud — uses services from multiple public cloud providers. The choice depends on security needs, cost, and flexibility.

2. Classify the various types of cloud services and compare them, highlighting key distinctions.

The three main service models: (1) IaaS (Infrastructure as a Service) — provides basic infrastructure like virtual servers and storage on a pay-as-you-go basis (e.g., AWS EC2). The user manages the operating system and applications. (2) PaaS (Platform as a Service) — provides a complete development and deployment environment (e.g., Google App Engine). The user only manages their code. (3) SaaS (Software as a Service) — provides ready-to-use hosted software (e.g., Google Workspace, Gmail). The user just uses the software. Key distinction: IaaS gives the most control, SaaS gives the least; PaaS is in the middle. As you move from IaaS to SaaS, the provider manages more, and the user manages less.

3. Discuss the advancements and benefits of edge computing in modern technology.

Edge computing has advanced with faster local processors, 5G networks, and AI chips. Its benefits: (1) Reduced latency — processing near the source enables real-time responses (autonomous vehicles, industrial automation). (2) Bandwidth savings — only summaries are sent to the cloud. (3) Reliability — works during network outages. (4) Privacy — sensitive data stays local. (5) Cost efficiency — less data transfer to central servers. Applications include smart cities, IoT devices, augmented reality, and content delivery. Edge computing complements cloud computing rather than replacing it.

4. Explain the concept of serverless architectures and their impact on application development.

Serverless architecture removes server management from application development. Developers write functions that run on demand in response to events (HTTP requests, database changes, file uploads). The cloud provider handles provisioning, scaling, and maintenance. Impact on development: (1) faster time-to-market — no infrastructure setup; (2) lower costs — pay per execution; (3) automatic scaling — handles traffic spikes effortlessly; (4) shift in mindset — developers think in functions and events rather than servers. Examples: AWS Lambda, Google Cloud Functions, Azure Functions. It is ideal for APIs, data processing, and event-driven apps.

5. Describe 'Cloud Deployment Models' with examples.

Cloud deployment models: (1) Public Cloud — shared infrastructure over the internet. Example: a small business using Google Drive for storage. (2) Private Cloud — dedicated to one organisation. Example: a bank running its own private cloud for sensitive data. (3) Hybrid Cloud — a mix of public and private. Example: a company keeps its database private but uses public cloud for its website. (4) Community Cloud — shared by organisations with similar needs. Example: several hospitals sharing a healthcare cloud. (5) Multi-Cloud — using multiple providers. Example: a company using AWS for computing and Google Cloud for AI. Each model balances cost, security, and control differently.

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